Taqrery PDF workflow

A practical review of how this project creates school reports and a step-by-step guide to producing a downloadable PDF.

What the project does

Taqrery is an Arabic-first school document app. Educators choose a teacher, counselor, principal, or vice-principal template; enter or load report data; preview the document; and download a PDF. The frontend is React 19 and TypeScript. Document pages use @react-pdf/renderer with Cairo font files and right-to-left layouts.

The PDF flow in the code

1. **Choose a template.** The template catalog routes the user to a section-specific form.
2. **Prepare data.** Form values and optional remote presets become a clean document model; each template keeps rendering logic in its document component.
3. **Render on demand.** The shared createReactPdfBlob function queues React PDF work, normalizes strings to NFC, rebuilds font state, and retries recoverable font failures.
4. **Use the blob.** The UI creates a preview URL, prints the blob, downloads it, or sends it to PDF.js for PNG/JPEG/ZIP export.
5. **Release resources.** Preview URLs are revoked when replaced or unmounted; browser download URLs are revoked after a delay.

> **Key distinction**
>
> The README recommends generating one blob and reusing it for all actions. In the current shared report route, each preview, print, or download handler calls createBlob() again. Both paths work, but reuse would avoid repeated rendering when form data has not changed.

Evidence reviewed: README.md, docs/pdf-generation-full-guidelines.md, shared report route, PDF engine, layout, preview, download, print, and image export modules.

Create your own PDF

Use these steps in the running Taqrery application.

In the browser

1. **Open a section.** From the template catalog, choose Teachers, Counselor, Principal, or Vice Principal.
2. **Select a report template.** Open the form that matches the document you want. Some templates can load preset content from configured JSON URLs.
3. **Fill and review the fields.** Enter the required school and report details. Optional blank values should not leave empty labels in the final PDF.
4. **Preview.** Use the Preview action. The app renders a PDF blob only when requested, then shows it in the preview area.
5. **Download PDF.** Use Download. The browser saves the PDF; Arabic names are normalized and sanitized by the shared download helper.
6. **Optional outputs.** Use Print, first-page PNG/JPEG, or all-pages image ZIP where the template offers these actions.

For local development

```
yarn install --frozen-lockfile
yarn dev
# Open the local URL printed by Vite, then follow the browser steps above.
# Before release: yarn validate
```

Template authoring path

When adding a new report type, put types.ts, data.ts, document.tsx, and index.ts in a template folder under src/pdf/react-pdf-templates/. Register Cairo fonts, use the shared PdfPageLayout for consistent branding, and call createReactPdfBlob from the route only after data is ready. The workspace component supplies the preview, download, print, share, and image-export controls.

> **How this report was made**
>
> This report was generated with the project's installed @react-pdf/renderer package and bundled Cairo fonts. It demonstrates the same document-rendering technology while remaining a standalone downloadable review artifact.

Where to look in the repo

Use this map when you need to change a template or troubleshoot an export.

Relevant files

Area	Responsibility
Template UI	src/routes/... and src/components/pdf-template/pdf-template-workspace.tsx
Document model	src/pdf/react-pdf-templates/<section>/<template>/types.ts and data.ts
PDF layout	document.tsx plus shared/pdf-layout.tsx and shared branding
Rendering	src/pdf/engines/react-pdf-engine.ts
Arabic fonts	shared/register-arabic-fonts.ts and public/fonts/Cairo-*.ttf
Download / print	src/pdf/print/download-blob.ts and print-pdf-blob.ts
Image export	src/pdf/image-export/export-pdf-images.ts and pdfjs.ts

Things to check before sharing a generated PDF

- Use the correct fixed page size (A4 or A3) and inspect page breaks, especially for long Arabic text and tables.
- Check that Cairo fonts load from the configured source; local files are the fallback for restricted or offline use.
- Confirm the content reflects the latest form values. The current shared report action renders separately on each request.
- Use yarn validate before a release. The command checks formatting, linting, types, tests, coverage, and the build.

> **Download behavior**
>
> In a browser, the helper creates a download link from the blob and revokes its URL after 60 seconds. In a React Native WebView, it sends the file through the native bridge; version 1 supports chunked transfer.

Scope: source review only. This document describes the implementation visible in the supplied repository; it is not a test result for every production template.